

NATURAL LANGUAGE PROCESSING (PRACTICE)

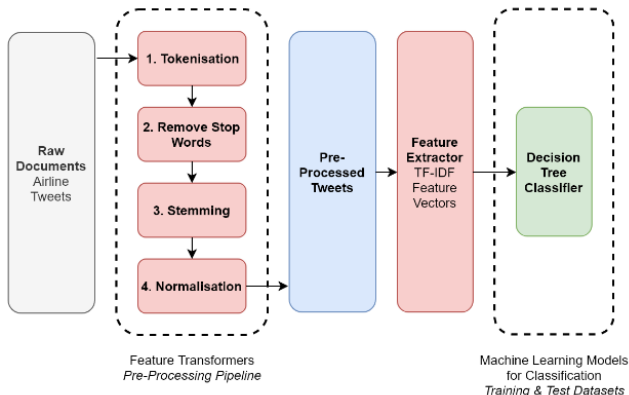
NLP 251 - Lab 2: Exploring and Preprocess Text Data



Department of Computer Science and Engineering
Ho Chi Minh University of Technology, VNU-HCM

Data Preprocessing

Data Preprocessing



- Data preprocessing is a crucial step
- 70%-80% of effort goes into data processing

Why is text preprocessing necessary?



- Multiple sources: web, HTML, documents...
- Contains noise and unclean data
- Goal: Convert to understandable format

Text Data Preprocessing Techniques

Text Data Preprocessing Techniques

- Convert to lowercase
- Remove punctuation
- Remove stop words
- Standardize text
- Correct spelling
- Tokenize
- Lemmatization and stemming
- Explore text data
- Build a preprocessing program

Converting to Lowercase

	A	B
1	Case	Example Text
2	Upper Case	THIS IS A SAMPLE TEXT
3	Lower Case	this is sample text
4	Proper Case	This Is Sample Text
5	Sentence Case	This is sample text

Problem: All text data needs consistent formatting, ensures "NLP" and "nlp" are treated the same

- Input: Text content
- Output: Text converted to lowercase
- Solution: Use `lower()` function in Python

Removing Punctuation

	A	B	C	D
	Text with Punctuation			Remove Punctuation
1				
2	"Apple"			Apple
3	(Pear). 5			Pear 5
4	{[Orange]}			Orange
5	Lemon;;; :::			Lemon
6	Lychee!			Lychee
7	<Blueberry>			Blueberry
8	Dash-test			Dashtest
9	TEST~!#\$%^&*(_+{} []""';<>?.,			TEST

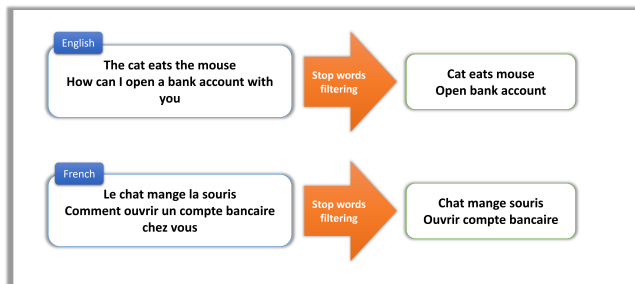
Target:

- Important because punctuation doesn't add information
- Reduces data size while improving computational efficiency

Problem

- Input: Text content
- Output: Text with punctuation removed
- Solution: Use regular expressions and replace() function in Python

Removing Stop Words



Stop words: common, but not very meaningful. Removing them helps to: reduce data size and potentially improve model performance.

- Solution:

- Use NLTK library
- Build a list of stop words, then use it to remove the stop words present in the text. (VNese Ex: <https://github.com/stopwords/vietnamese-stopwords>)

Standardizing Text

Raw	Normalized
2moro	tomorrow
2mrrw	tomorrow
2morrow	tomorrow
tomrw	tomorrow
b4	before
otw	on the way
:)	smile
:-)	smile
;-)	smile

- Input: Text content
- Output: Standardized text
- Solution: Create dictionary to look up abbreviations and short forms

Correcting Spelling

Common Examples

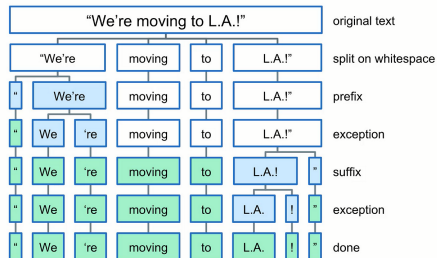
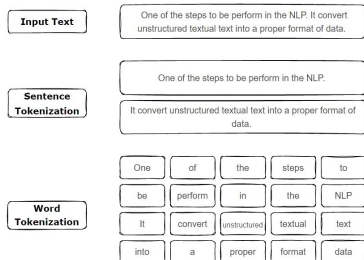
studying → studying
intresting → interesting
aquire → acquire

- Text contains spelling errors (user evaluations, blogs, tweets, etc.)
- Reduce the number of duplicates of words. For example, if not fixed, "studing" and "studying" would be considered two different words.

Problem:

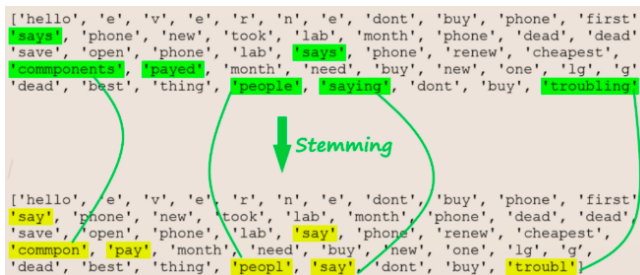
- Input: A text containing content.
- Output: The text with corrected spelling errors.
- Solution: Use **TextBlob** library

Tokenizing



- Input: Text content
- Output: Separated sentences or words
- Solutions:
 - For English: NLTK, SpaCy, TextBlob
 - For Vietnamese: VnCoreNLP, underthesea, coccoc-tokenizer

Stemming and Lemmatization



Stemming

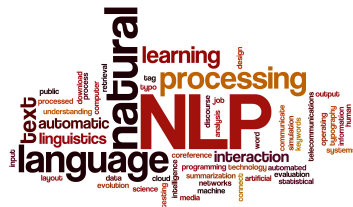
- Input: "fishing", "fishes"
- Output: Root "fish"
- Removes prefixes/suffixes

Lemmatization

- Input: "good", "best"
- Output: Root "good"
- Considers semantics

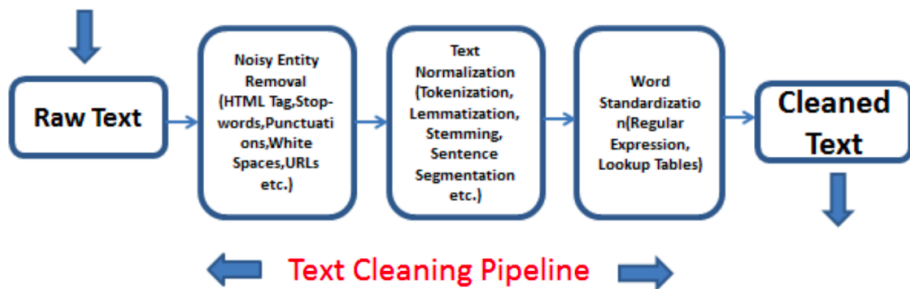
Solution: Use **NLTK** or **TextBlob**

Exploring Text Data



- **Input:** Text content
- **Output:**
 - Word count
 - Word frequency
 - Distribution of words longer than 3 characters
 - Word cloud
- **Solution:** Use NLTK or TextBlob

Building Preprocessing Program



- **Input:** Text content
- **Output:** Preprocessed text
- **Solution:** Create preprocessing function incorporating all techniques discussed

THANKS FOR
LISTENING!